Homework 2

Individual assignments due in Canvas on Tuesday, 22 September

Group collaborative key due on Tuesday, 29 September

1. On Homework #1 you worked with the data in problem 15, regarding age (in months) of unassisted walking among a sample of 240 children. In that problem, you found the mean, SE, and a 95% confidence interval for average age of walking. In that problem, you were not told the size of the population and so did not account for the fpc. It turns out the data were collected from a pediatric outpatient clinic that serves 5850 children.

   a. Define/describe the following terms for this problem:
      i) Objective of the study
      ii) Target population
      iii) Observation unit
      iv) Population size
      v) Characteristic of interest
      vi) Sample unit
      vii) Sample size
      viii) Sample design
      ix) Number of possible samples for this design (it's okay to leave as a function of factorials)
      x) Inclusion probability
      xi) Sampling weight
      xii) fpc
   b. Use hand calculations to adjust the SE and 95% confidence intervals for the fpc.
   c. Use `svymean` in R to verify the values from (b).
   d. Do you think the fpc can be ignored for this problem? Explain your reasoning.

2. Book problems from Chapter 2: #13 (objects in bins; #36 in my online book; on part (a), use the seed=91811), #18 (SSC; #27 in my online book)